Codebusters

Spies, soldiers, and computers all hide messages in codes. In Codebusters you learn ten kinds of ciphers and the tricks for cracking them, then race your partner against the clock.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 2	Crack the Code: Shifts and Mirrors
Lesson 2	Fri Nov 6	Taps and Pens: Codes You Must Memorize
Lesson 3	Fri Dec 11	Aristocrats: Letter Detectives
Lesson 4	Fri Jan 29	Templar Keys and No-Hint Aristocrats
Lesson 5	Fri Feb 26	Mock Tournament: Codebusters Test
Lab I1	Before Lesson 2	Vigenere: A Different Shift for Every Letter (home or an open Friday)
Lab I2	Before Lesson 3	Baconian: Codes Made of A and B (home or an open Friday)
Lab I3	Before Lesson 4	Galactic Alphabet: Read the Secret Sentence (home or an open Friday)

At the tournament

A 50-minute written test with only pencils: partners decode as many secret messages as they can, using a resource sheet (letter frequencies, Atbash, Galactic Alphabet (as the sentence THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG), Baconian and Vigenere tables) but NO PigPen, Tap Code, or Knight's Templar keys, so those must be memorized.

Keep every handout for this event behind this divider, in order. Bring this binder to every Field A Fridays meeting.

Codebusters: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can decode a Caesar (shift) cipher that moves letters up to 3 places forward or back.	L1 L5	☐	☐
I can decode an Atbash cipher by writing A to Z over Z to A.	L1 L5	☐	☐
I can explain how the test is scored: 2 or fewer mistakes on a question still gets full points, and each extra mistake loses 50.	L1 L5	☐	☐
I can draw the Tap Code grid from memory (C and K share one box) and turn number pairs into letters.	L2 L4 L5	☐	☐
I can draw the PigPen key from memory and decode PigPen symbols.	L2 L4 L5	☐	☐
I can count letters and use letter frequency (E, T, A, O, I, N are most common) to guess what a cipher letter stands for.	L3 L4	☐	☐
I can use short words (A, I, THE, AND) and word patterns to solve an Aristocrat that has a hint.	L3 L5	☐	☐
I can solve an Aristocrat with no hint, remembering that no letter ever stands for itself.	L3 L4	☐	☐
I can decrypt a Vigenere cipher when I am given the key word.	L5 I1	☐	☐
I can decode a Baconian cipher by sorting up to 4 symbols into an A group and a B group, then using the table.	L5 I2	☐	☐
I can draw the Knight's Templar key from memory and decode Templar symbols.	L4 L5	☐	☐
I can label THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG on the resource sheet and use it to decode Standard Galactic Alphabet symbols (Y is always ||).	L5 I3	☐	☐
I can split a timed test with my partner and pick the questions we can finish first.	L5	☐	☐

Crack the Code: Shifts and Mirrors

TODAY I CAN

- I can decode a Caesar (shift) cipher that moves letters up to 3 places forward or back.
- I can decode an Atbash cipher by writing A to Z over Z to A.
- I can explain how the test is scored: 2 or fewer mistakes on a question still gets full points, and each extra mistake loses 50.

A cipher is a secret way to write a message. Today you crack two of the ten ciphers on the test: the Caesar shift and Atbash.

Today's steps: run it yourselves (22 minutes)

You need: Paper strips (2 per student), Lesson 1 race cards (3), Lesson 1 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading Words to know out loud. Tell your partner why 2 small mistakes are OK, but you should still check your work.
2. Both 5 min. Set the timer for 5 minutes. Each of you makes a Caesar strip: write A to Z on two strips so the letters line up. Slide the bottom strip so D sits under A (shift +3), then so X sits under A (shift -3).
3. Event Captain 8 min. Set the timer for 8 minutes. The Event Captain gives each pair a race card. Use How to crack a shift: try all 6 shifts on the shortest word. When a pair finishes, the Captain checks it on the answer card and hands over the next card.
4. Both 6 min. Set the timer for 6 minutes. Fold a strip in half. Write A to M on top and Z to N under it (Z under A). Decode Cipher 3 (Atbash). Done early? Decode Cipher 1 and Cipher 2.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Check Ciphers 1 to 3 and fix any letter you missed in a different color.

Words to know

Plaintext	The real message.
Ciphertext	The scrambled message you see.
Caesar shift	Every letter moves the same number of places. On our test the shift is 1, 2, or 3, forward or back.
Atbash	The alphabet is flipped: A=Z, B=Y, C=X, and so on.
Scoring	2 or fewer mistakes on a question still gets full points. Each extra mistake costs 50 points.

How to crack a shift

1. Find the shortest word.
2. Try moving its letters back 1, 2, or 3, then forward 1, 2, or 3.
3. When it turns into a real word, you found the shift.
4. Use the same shift on every letter.

Cipher 1 (Caesar)

Hint: Each letter moved 2 places forward.

E Q F G U J G N R U R K G U U J C T G U G E T G V U

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Cipher 2 (Caesar)

Hint: Find the shift yourself (try the word ZXK). Shift that made it: _____

Y B B P Z X K P B B R I Q O X S F L I B Q I F D E Q

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Cipher 3 (Atbash)

L D O H X Z M M L G N L E V G S V R I V B V H

_ _ _ _

Taps and Pens: Codes You Must Memorize

TODAY I CAN

- I can draw the Tap Code grid from memory (C and K share one box) and turn number pairs into letters.
- I can draw the PigPen key from memory and decode PigPen symbols.

The test will NOT give you the Tap Code or PigPen keys. You must draw them from memory.

Today's steps: run it yourselves (22 minutes)

You need: 4 index cards, PigPen key card, PigPen message card, Lesson 2 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Read the Tap Code box. Each of you draws the 5 by 5 grid on scrap paper: ABCDE, FGHIJ, LMNOP, QRSTU, VWXYZ. C and K share a box.
2. Both 5 min. Set the timer for 5 minutes. Decode Tap Message 1 and Tap Message 2. If a box could be C or K, pick the letter that makes a real word.
3. Both 5 min. Set the timer for 5 minutes. Copy the PigPen key card into the Draw the PigPen key box. Each symbol is the shape of the lines around that letter's space, plus the dot if it has one. Then decode the PigPen message card.
4. Both 6 min. Set the timer for 6 minutes. Do Read the Shapes. Then each of you writes a 3 to 5 letter word in PigPen on an index card and swaps with your partner to decode.
5. Both 2 min. Set the timer for 2 minutes. Flip the handout and key card face down. Draw both keys on your whiteboard from memory, then check your partner's.

Check your work: Ask your Event Captain for the Lesson 2 answer card and check both Tap Messages, the message card, and Read the Shapes. Check your memory keys against the key card and fix mistakes in a different color.

Tap Code

The grid is 5 by 5. Rows: ABCDE / FGHIJ / LMNOP / QRSTU / VWXYZ. C and K share one box, so there is no K. First number = row (down), second = column (across): 2 3 = H.

Tap Message 1 (/ means a new word)

1	2	3	4	5	6	7	8	9	10	11	12
4 3	3 5	1 1	1 3	1 5	/	4 2	3 4				
					/						
1 3	1 3	4 3									

Tap Message 2

1	2	3	4	5	6	7	8	9	10	11	12
2 1	4 2	3 4	2 2	4 3	/	1 4	4 2				
					/						
2 4	3 3	1 3	/	4 4	2 3	4 2	3 4				
			/								
4 5	2 2	2 3	/	4 3	1 3	2 4	3 3				
			/								

Draw the PigPen key (two grids and two X's)

Read the Shapes: write the letter for each PigPen symbol to spell a word

Symbol 1	Symbol 2	Symbol 3	Symbol 4
Lines on right, bottom, left. No dot.	Lines on top, bottom, left. Dot.	Lines on all 4 sides. Dot.	Lines on all 4 sides. No dot.

At home before next time: Do Independent Lab I2 (Baconian) before Lesson 3 on Dec 11. Also draw the Tap Code grid and PigPen key from memory 3 times, then check them. Goal: both in 1 minute. Best time: _____. Case File: open Case 2, Operation Knock-Knock (p. 7), decode the Tap Code and PigPen messages, and tap a word to someone at home to earn the Tap Talker and PigPen Pro badges.

Aristocrats: Letter Detectives

TODAY I CAN

- I can count letters and use letter frequency (E, T, A, O, I, N are most common) to guess what a cipher letter stands for.
- I can use short words (A, I, THE, AND) and word patterns to solve an Aristocrat that has a hint.
- I can solve an Aristocrat with no hint, remembering that no letter ever stands for itself.

An Aristocrat swaps every letter for a different letter. There is no simple rule, so you solve it like a detective: count letters, find short words, and test guesses.

Today's steps: run it yourselves (27 minutes)

You need: Sticky notes, Lesson 3 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Take turns reading the intro and Detective tricks out loud. Write E T A O I N S H R D L U at the top of your whiteboard.
2. Both 5 min. Set the timer for 5 minutes. Count how many times each letter shows up in Cipher A and fill in Count the top letters. Guess what the most common letter really is.
3. Both 5 min. Set the timer for 5 minutes. Hint: Cipher A starts with THE. Write T, H, and E above every matching cipher letter in the whole message. Then answer the question about FGNV.
4. Both 10 min. Set the timer for 10 minutes. Finish Cipher A, then start Cipher B. Write guesses lightly, and keep a key strip (cipher letter on top, real letter under) so you never use one real letter twice.
5. Team 3 min. Set the timer for 3 minutes. Each pair tells the other pair one trick that helped them most.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Check your letter counts, Cipher A, Cipher B, and the FGNV question. Fix any you missed in a different color.

Detective tricks

Most common letters	E T A O I N S H R (E is the most common).
Short words	One letter: A or I. Common: THE, AND, IS, IT, TO, OF.
Word pattern	Letters that repeat, like the two L's in BELL (pattern 1-2-3-3).
Golden rule	No letter ever stands for itself.

Cipher A

Hint: The first word is THE.

G T M F S H A F N F G N V N H K A G A F G T M L C B F M F G

— —

F G N V G B M N V G T

— — — — — — — — — — —

Count the top letters in Cipher A

Cipher letter	How many times	I think it is
G		
F		
N		
M		

Cipher B

Hint: The last word is IDEAS.

X B E J Q K E X K X R X Y S O J X K E L Q X R Q V K J X K E V J R

— —

X

—

1. The cipher word FGNV appears twice in Cipher A. What real word is it?

At home before next time: Before Lesson 4 on Jan 29: make an index card with the 9 most common letters (E T A O I N S H R) and the one-letter words (A, I), and quiz yourself until you know them. Finish Cipher B if you did not. Once a week, draw the Tap Code grid and PigPen key from memory and time yourself; Lesson 4 starts with a memory race. Do Independent Lab I3 (Galactic Alphabet) before Lesson 4: you will learn to read the Galactic sentence on the resource sheet. Case File: open Case 3, Operation Letter Count (p. 8), solve the Aristocrat with the THE hint and count letters on a book page to earn the Letter Detective badge.

Templar Keys and No-Hint Aristocrats

TODAY I CAN

- I can draw the Tap Code grid from memory (C and K share one box) and turn number pairs into letters.
- I can draw the PigPen key from memory and decode PigPen symbols.
- I can count letters and use letter frequency (E, T, A, O, I, N are most common) to guess what a cipher letter stands for.
- I can solve an Aristocrat with no hint, remembering that no letter ever stands for itself.
- I can draw the Knight's Templar key from memory and decode Templar symbols.

The test does NOT give you the Knight's Templar key, so you must memorize it, just like PigPen and Tap Code. Then you and your partner race the timer on the hardest kind of Aristocrat: one with no hint at all.

Today's steps: run it yourselves (27 minutes)

You need: Knight's Templar key sheets (1 per student), Templar message card, Lesson 4 answer card. Set the timer for each step.

1. Both 2 min. Set the timer for 2 minutes. Read the intro and How to memorize a key out loud. Look at the key sheet and find the frame shapes and where the dots go.
2. Both 4 min. Set the timer for 4 minutes. Each of you copies the key into Draw the Knight's Templar key, saying each letter out loud as you draw it.
3. Both 4 min. Set the timer for 4 minutes. Turn the key sheets face down. Partner A draws the whole key from memory on scrap paper while Partner B checks it with the sheet. Then switch. Fix mistakes in a different color.
4. Both 3 min. Set the timer for 3 minutes. Ask the Event Captain for the Templar message card. Decode it from memory, then check with the key sheet. Write it in the first question.
5. Event Judge 4 min. Memory race: key sheets face down. The Judge says 'Go' and counts seconds out loud. Everyone draws PigPen, Tap Code, and Templar and writes each time in Memory race. Partners check each other's keys and write the mistakes.
6. Both 3 min. Set the timer for 3 minutes. Count the letters in the No-hint Aristocrat and fill in Count the top letters. Which cipher letters might be E, T, A, or S?
7. Both 7 min. Set the timer for 7 minutes and race it on the No-hint Aristocrat. Partner A keeps a key strip, Partner B tests words; switch jobs after 3 minutes. Start with BRR: what 3-letter word has a double letter? Answer the last question.

Check your work: Ask your Event Captain for the Lesson 4 answer card. Check the Templar card, your letter counts, the Aristocrat, and the BRR question. Fix any you missed in a different color.

How to memorize a key

Draw the frame first, then write the letters in order in the spaces. Each letter's symbol is the shape of the lines around its space, plus a dot if its space has one. Draw it from memory once a day until it takes less than 1 minute.

Draw the Knight's Templar key

Memory race

Key	Time (seconds)	Mistakes
PigPen		
Tap Code		
Knight's Templar		

No-hint Aristocrat

```
B R R   V S C G M N C   J B U G   C V O   R G E C   B S X   N J L G G   F Q X

_ _ _   _ _ _ _ _ _ _   _ _ _ _   _ _ _   _ _ _ _   _ _ _   _ _ _ _ _   _ _ _

I   Y B L N C

_   _ _ _ _ _
```

Count the top letters

Cipher letter	How many times	I think it is
C		
G		
B		
R		
N		

Detective tricks

Double letters	Common ones: LL, EE, SS, OO, TT
Golden rule	No letter ever stands for itself.

1. What does the Templar message card say?

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2. What real word is BRR, and how did you know?

At home before next time: Before Lesson 5 on Feb 26 (the mock test): draw the Knight's Templar key, PigPen key, and Tap Code grid from memory 3 times each, and write your best times in Memory race. Goal: each under 1 minute with 0 mistakes. Finish the No-hint Aristocrat if you did not. The mock has every cipher type, so also redo one part of Lab I1, I2, or I3 (Vigenere, Baconian, or Galactic). Bring a sharp pencil and an eraser. Case File: open Case 4, Operation No Hint (p. 9), solve the no-hint Aristocrat and log your key times on the Memory Race page (p. 11) to earn the No-Hint Hero badge.

Mock Tournament: Codebusters Test

TODAY I CAN

- Everything on my can-do checklist (the page right after the divider). Today is the time to show it.

This is a practice tournament. You have 20 minutes. Split the questions with your partner and do the ones you can finish first. Points are in parentheses.

Today's steps: run it yourselves (27 minutes)

You need: 3 symbol cards per pair (Q7 to Q9), Resource sheets (1 per pair), Lesson 5 answer card. Set the timer for each step.

1. Event Judge 1 min. Read the intro out loud: 20 minutes, split the questions, any order, binders closed except the resource sheet. Hand each pair the 3 symbol cards (Q7 to Q9).
2. Both 1 min. Set the timer for 1 minute. Look over the test with your partner and decide who starts which question.
3. Event Judge 20 min. Set the timer for 20 minutes and say 'Go.' Call out time left at 10, 5, and 1 minute. Everyone works the test with their partner; no help between pairs.
4. Both 5 min. Set the timer for 5 minutes. Swap papers with the other pair. Use the answer card to count mistakes on each question and fill in Score it: 2 or fewer mistakes is full points, each extra mistake loses 50, never below 0.

Check your work: Give the papers back and look at Score it together. Circle the question type that cost your pair the most points; that is your at-home practice.

Q1 Caesar (100)

Hint: Shift of 3 or less, either way.

L Z F M D S R O T K K N M H Q N M

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Q2 Atbash (100)

K O Z M G H M V V W O R T S G

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Q3 Aristocrat (250)

Hint: The word SOLID is in the message.

O B J N S D B Q G N B V Z X F W B X F P E F W Z S B L B V

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Q4 Vigenere (200)

Hint: The key is ICE.

O N E K K I Z U G I T Z M X E T N I G U

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Q5 Tap Code (150)

1	2	3	4	5	6	7	8	9	10	11	12
1 2	3 1	1 1	1 3	1 3	/	2 3	3 4				
					/						
3 1	1 5	4 3									

Q6 Baconian (150)

Group	1	2	3	4
Symbols	@****	+++++	*#+*+	*@@**
A or B				
Letter				

Q7 to Q9: symbol cards from your coach (150 each)

Question	Answer
Q7 PigPen	
Q8 Knight's Templar	
Q9 Galactic Alphabet	

Score it

Question	Mistakes	Points
Q1		/100
Q2		/100
Q3		/250
Q4		/200
Q5		/150
Q6		/150
Q7 to Q9		/450
Total		/1400

At home before next time: Look at your score table. Pick the 2 cipher types that cost you the most points and redo that part of Lab I1, I2, or I3, Lesson 3, or Lesson 4, or ask Coach for that part of Practice Test 1. Draw all three memory keys one last time the night before the tournament. Case File: do Case 5, Operation Final Countdown (p. 10), a 15-minute mini test, and score it to earn the Speed Decoder badge.

Lab I1: Vigenere: A Different Shift for Every Letter

THIS LAB HELPS ME

- I can decrypt a Vigenere cipher when I am given the key word.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. A Vigenere cipher is like many Caesar shifts in a row. A key word tells you a different shift for each letter. The test gives you the key word, and you undo the shifts.

Steps (29 minutes)

You need: Pencil and eraser, 1 index card, Optional: Vigenere square from the Open Lab bin. Set the timer for each step.

1. Me 4 min. Set a timer for 4 minutes. Read the intro and Many shifts in a row. In Letter numbers for Vigenere, find the numbers for S, U, and N and write them in the margin.
2. Me 5 min. Set the timer for 5 minutes. In Vigenere 1, write SUNSUN under HFNFNF. H is 7 and S is 18: 7 minus 18 is -11, add 26 to get 15, which is P. Finish the word (it should be PLANTS).
3. Me 7 min. Set the timer for 7 minutes. Finish Vigenere 1 with How to decrypt a Vigenere. Keep going with the key (S U N S U N…) and skip the spaces.
4. Me 7 min. Set the timer for 7 minutes. Decrypt Vigenere 2 (key BEE). Working with a partner? One writes the key letters, the other does the math, then switch.
5. Me 6 min. Set the timer for 6 minutes. Read Make your own. Write a 3 to 5 word message on an index card in Vigenere with the key CAT. Write the real message on the back. Then answer the question at the bottom.

Check your work: A real sentence means you did it right. To check your CAT card, decrypt it yourself (subtract the key) and see if you get your message back. Then check with the Lab I1 answer card in the Open Lab bin, or at the next Huddle.

Many shifts in a row

Each key letter is a shift: A means shift 0, B means shift 1, C means shift 2, and so on. With the key SUN, the 1st letter shifts 18, the 2nd shifts 20, the 3rd shifts 13, and then the key starts over.

Letter numbers for Vigenere

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

How to decrypt a Vigenere

1. Write the key word under the ciphertext again and again (skip spaces).
2. Change both letters to numbers.
3. Cipher number minus key number. Below 0? Add 26.
4. Change the answer back to a letter.

Vigenere 1 (key: SUN)

H F N F N F E U X W Z B G X S J I Z D C T Z N

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Vigenere 2 (key: BEE)

C I I T H E O G I U S W I S A X L I S I J P S H J W

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Make your own (key: CAT)

1. Write your message. Write CATCAT... under it (skip spaces).
2. Change both letters to numbers. C is 2, A is 0, T is 19.
3. Real number PLUS key number. 26 or more? Subtract 26.
4. Change the answer back to a letter. Example: DOG becomes FOZ.

1. Why is a Vigenere harder to crack than a Caesar shift?

Bring to the next meeting: Bring your CAT message card to Lesson 2. Your partner decodes it in the Do Now.

Lab I2: Baconian: Codes Made of A and B

THIS LAB HELPS ME

- I can decode a Baconian cipher by sorting up to 4 symbols into an A group and a B group, then using the table.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Baconian uses a table from the test's resource sheet. Every letter turns into a group of 5 symbols of only two kinds, an A kind and a B kind.

Steps (27 minutes)

You need: Pencil and eraser. Set the timer for each step.

1. Me 4 min. Set a timer for 4 minutes. Read the Baconian table. Every letter is a group of 5 made of only A and B. Find D and E and say them out loud.
2. Me 7 min. Set the timer for 7 minutes. Do Baconian 1 using the hint. Write A or B under each symbol, then find each group in the table.
3. Me 6 min. Set the timer for 6 minutes. Do Baconian 2. Sort the 4 symbols into an A team and a B team and test it: if every group makes a real letter, you sorted them right. If not, swap A and B. Answer the question.
4. Me 7 min. Set the timer for 7 minutes. Make your own: write your first name in Baconian, but use two shapes of your own (like circles for A and squares for B) instead of the letters A and B. Trade with your partner or someone at home and decode theirs.
5. Me 3 min. Set the timer for 3 minutes. Look at the table and answer: which letters share a code? (Hint: look at I/J and U/V.)

Check your work: Every group should make a letter and the letters should spell a word. Then check with the Lab I2 answer card in the Open Lab bin, or at the next Huddle.

Baconian table (24 letters: I and J share, U and V share)

Letter	Code	Letter	Code
A	AAAAA	N	ABBAA
B	AAAAB	O	ABBAB
C	AAABA	P	ABBBA
D	AAABB	Q	ABBBB
E	AABAA	R	BAAAA
F	AABAB	S	BAAAB
G	AABBA	T	BAABA
H	AABBB	U/V	BAABB
I/J	ABAAA	W	BABAA
K	ABAAB	X	BABAB
L	ABABA	Y	BABBA
M	ABABB	Z	BABBB

Baconian 1 (hint: * and + stand for A. # and @ stand for B.)

Group	1	2	3	4	5
Symbols	+*+#*	*@#*#	*@+##	*+#**	#+*#*
A or B					
Letter					

Baconian 2 (no hint: sort the symbols yourself)

Group	1	2	3	4
Symbols	*+@*#	@***+	*##*@	++@@+
A or B				
Letter				

1. How did you know which symbols meant A in Baconian 2?

The sentence in code (write THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG under it)

N Z C K V R Y M T W P U A I P O S V E H L P G C W N Z C F D X Q

— — — — — — — — — — — — — — — — — — — — — — — — — — — — — — — —

B P J

— — —

Make your own: my first name in Baconian (my own 2 shapes)

Bring to the next meeting: Bring this lab to Lesson 3. The Team Captain will ask to see your Baconian answers and your Baconian name.

Lab I3: Galactic Alphabet: Read the Secret Sentence

THIS LAB HELPS ME

- I can label THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG on the resource sheet and use it to decode Standard Galactic Alphabet symbols (Y is always ||).

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. The Standard Galactic Alphabet (Galactic for short) is a secret alphabet made of symbols instead of letters. Every letter always has the same symbol, so once you crack a symbol, you know it everywhere in the message.

Steps (30 minutes)

You need: Pencil and eraser. Set the timer for each step.

1. Me 3 min. Set a timer for 3 minutes. Read How it works on the test and Clues that help. Find the symbol for Y (||) in the sentence. Which word is it in?
2. Me 5 min. Set the timer for 5 minutes. Step 1: say THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG out loud and write one letter in each box under the sentence. Then check: T, H, E, O, U and R appear more than once, and their symbols must match every time.
3. Me 3 min. Set the timer for 3 minutes. Study Look-alike symbols. Cover the letters with your hand and say each one.
4. Me 3 min. Set the timer for 3 minutes. Read the Example (STAR). Point to where each of its 4 symbols is in your labeled sentence.
5. Me 3 min. Set the timer for 3 minutes. Do the Look-alike quiz. Look closely: count the dots and the lines.
6. Me 4 min. Set the timer for 4 minutes. Do Practice 1. For each symbol, find it in your labeled sentence and write its letter in the box.
7. Me 6 min. Set the timer for 6 minutes. Do Practice 2, a quote like the one on the real test. Start with the short words, then use what makes sense.
8. Me 3 min. Set the timer for 3 minutes. Write your first name in Galactic in the last box. Trade with your partner or someone at home and decode theirs.

Check your work: Your labeled sentence should have the same letter under every matching symbol, and Practice 1 and 2 should be real words. Then check with the Lab I3 answer card in the Open Lab bin, or at the next Huddle.

How it works on the test

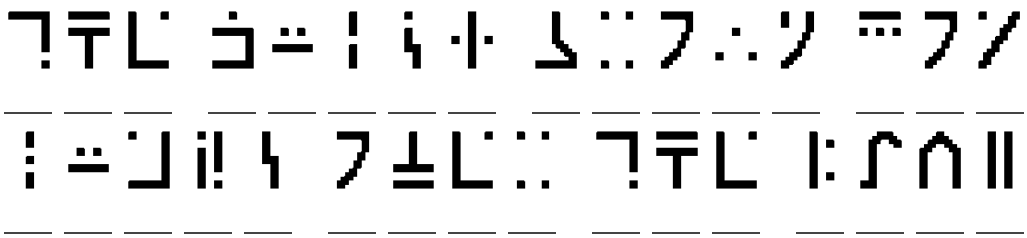
The resource sheet does NOT give you an A to Z chart. It gives you one sentence written in Galactic: THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG. That sentence uses all 26 letters, so it is your whole key. First you write the letters under the sentence, then you look up each symbol of the secret message. A Galactic question is worth about 80 points and is usually a 40 to 50 letter quote from a famous person.

Clues that help

Y is always ‖	Two straight lines. It is the 4th symbol of LAZY. Find it first.
Repeat letters	T, H, E, O, U and R are in the sentence more than once. Matching symbols prove you labeled it right.
Same symbol, same letter	A symbol never changes inside a message, so fill in every copy of a symbol as soon as you know it.
Word spaces stay	You can still see short words. A 1-symbol word is A or I. A 3-symbol word is often THE or AND.

Step 1: Label the sentence (write one letter under each symbol)

On the test, do this first on the resource sheet. It takes about 2 minutes and saves time on every question.

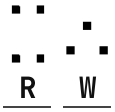


Look-alike symbols (go slow on these)



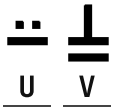
I J Y K L P

Up-and-down lines. I = one line with a gap. J = three pieces. Y = two full lines. K = a dot on each side. L = two dots on the right. P = ! next to an upside-down !.



R W

Only dots. R = 4 dots (a square). W = 3 dots (a triangle).



U V

U = two dots over a line. V = a post standing on a line, with a second line under it.



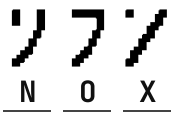
E M

Mirror twins. E = the corner is bottom left, dot on the right. M = the corner is bottom right, dot on the left.



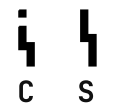
H T F Q

A bar on top. H = two bars and a stem. T = a bar with a line down the right and a dot under it. F = a bar with 3 dots. Q = a dot over a backwards C.



N O X

Slanted lines. N = a short line plus a slant. O = a bar on top that bends into a slant, like a 7. X = a dot plus one long slant.



C S

Zigzag lines. C has a dot on top. S does not.

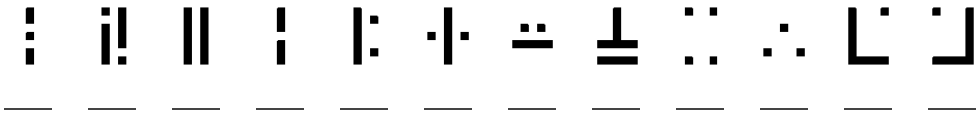
Example: STAR (solved for you)

S is the 5th symbol of JUMPS. T is the 1st symbol of THE. A is the 2nd symbol of LAZY. R is the 2nd symbol of BROWN.



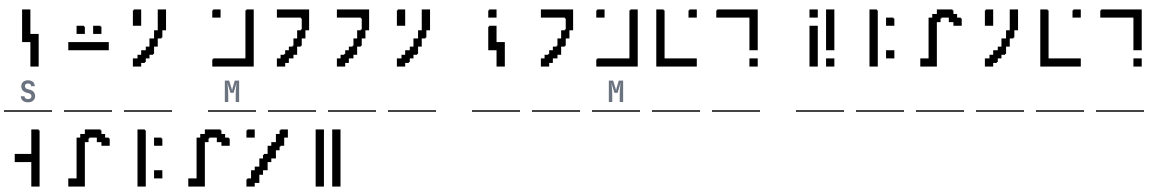
S T A R

Look-alike quiz: write the letter for each symbol



Practice 1: space words

Some letters are filled in to get you started.



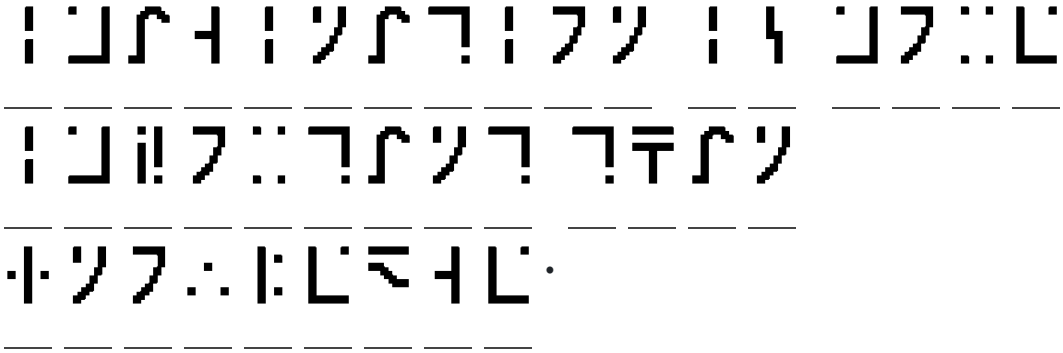
S M M

Practice 2: space words

Some letters are filled in to get you started.

M

Practice 2: Solve this quote from Albert Einstein which has been encoded with the Standard Galactic Alphabet.



My first name in Galactic (draw the symbols)

Bring to the next meeting: Bring this lab to Lesson 4. The Team Captain will ask to see your labeled sentence and the Einstein quote.